

BaS Furthest-From-Opportunity Model: Technical Details

This document describes the OLS regression model used to predict the share of selected BaS scholars who are 'furthest from opportunity' (both first-generation college students AND from families earning at or below 65% of the HUD Median Family Income).

Model Specification

Furthest Share (Selected) = $B_0 + B_1 * \ln(N \text{ Furthest Applicants} / N \text{ Slots}) + \text{error}$

The natural log transform is used because the relationship exhibits diminishing returns: when there are already many furthest applicants per slot, adding more has a smaller marginal effect on the selected share.

Fitted Parameters

Intercept (B_0): 0.579035

Slope (B_1): 0.180109

R-squared: 0.8139

p-value (slope): 0.000025

Number of observations: 13

Residual Std Error: 0.031582

Inverse Formula (Target Setting)

To find the number of furthest-from-opportunity applicants needed:

$N \text{ Furthest Needed} = N \text{ Slots} \times \exp((\text{Target Share} - B_0) / B_1)$

$N \text{ Furthest Needed} = N \text{ Slots} \times \exp((\text{Target Share} - 0.5790) / 0.1801)$

Good / Better / Best Estimates

Good: The number of furthest-from-opportunity applicants needed so that the best-fit regression line predicts the target share. This is the minimum needed if the historical relationship holds exactly.

Best: The number of furthest-from-opportunity applicants needed so that the lower bound of the 95% prediction interval equals the target share. This provides a buffer against year-to-year variability. Even in an unfavorable year, you would still expect to meet the target.

Better: The midpoint between Good and Best.

Prediction Intervals

The 95% prediction interval accounts for both estimation uncertainty (how well we know the regression line) and residual variance (natural year-to-year scatter). The interval width varies with the predictor value, being narrowest near the center of the observed data and wider at the extremes.

Historical Data

Year	Applicants	Selected	Furthest	Share	Ratio
2013-14	985	778	399	42.4%	0.51
2014-15	917	782	300	38.2%	0.38
2015-16	1,242	1,057	469	43.8%	0.44
2016-17	2,479	1,450	984	52.5%	0.68
2017-18	2,503	1,860	998	52.9%	0.54
2018-19	2,966	1,862	912	42.7%	0.49
2019-20	3,160	1,850	1,122	48.7%	0.61
2020-21	2,573	750	917	56.9%	1.22
2021-22	2,306	750	786	60.4%	1.05
2022-23	1,480	1,262	503	39.5%	0.40
2023-24	1,522	1,275	555	41.9%	0.44
2024-25	1,682	1,275	647	48.3%	0.51
2025-26	2,253	1,275	839	53.6%	0.66

Caveats

- The model is fit on 13 observations (one per application year). Standard errors should be interpreted cautiously.
- The model assumes the program's selection preference for furthest scholars remains approximately stable over time.
- Extreme target values (near 0% or 100%) produce unreliable predictions.
- The applicant pool composition is exogenous. The model indicates how many furthest applicants are needed but does not control whether they apply.